Local Bike Final Case

Analytics Engineering

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Summary

1. Define the analysis axes
2. Data modeling
3. Implementing tests and documentation
4. Visualization and sharing

1. Define the analysis axes

Objective

Define key analysis axes to support operations.

Provide strategic insights to **optimize processes, maximize sales and increase revenue.**

Local Bike context

A mission to democratize cycling in the United States (cycling infrastructure and culture, soft mobility, health, carbon footprint). Strong community and a focus on quality over quantity.

1. The 3 selected analysis axes

Selected from other axes identified in the available data

1

Revenue by store, category and brand

Identify winning categories and brands to better allocate resources (promotions, staff, events, stock, etc.).

2

Stock analysis

- Products never sold: promote or discontinue.
- Overstock: trigger promotions.
- Stockouts: rebalance or restock.

3

Top 10 cities by volume sold (last year)

- Organize local events to strengthen the community.
- Obtain sponsorship from popular brands.

Expected impact: better resource allocation, higher sales and revenue, a stronger community.

2 & 3. (dbt) Modeling and implementation

3-layer architecture

Staging (views)

Cleaning and standardization of source tables. Personal data columns of customers and staff were removed from the original dataset.

Intermediate (views)

`int__production`, `int__sales`: enrichment and joins, organized by the Production / Sales split in the brief.

Mart (tables)

Analysis-ready models, one per axis.

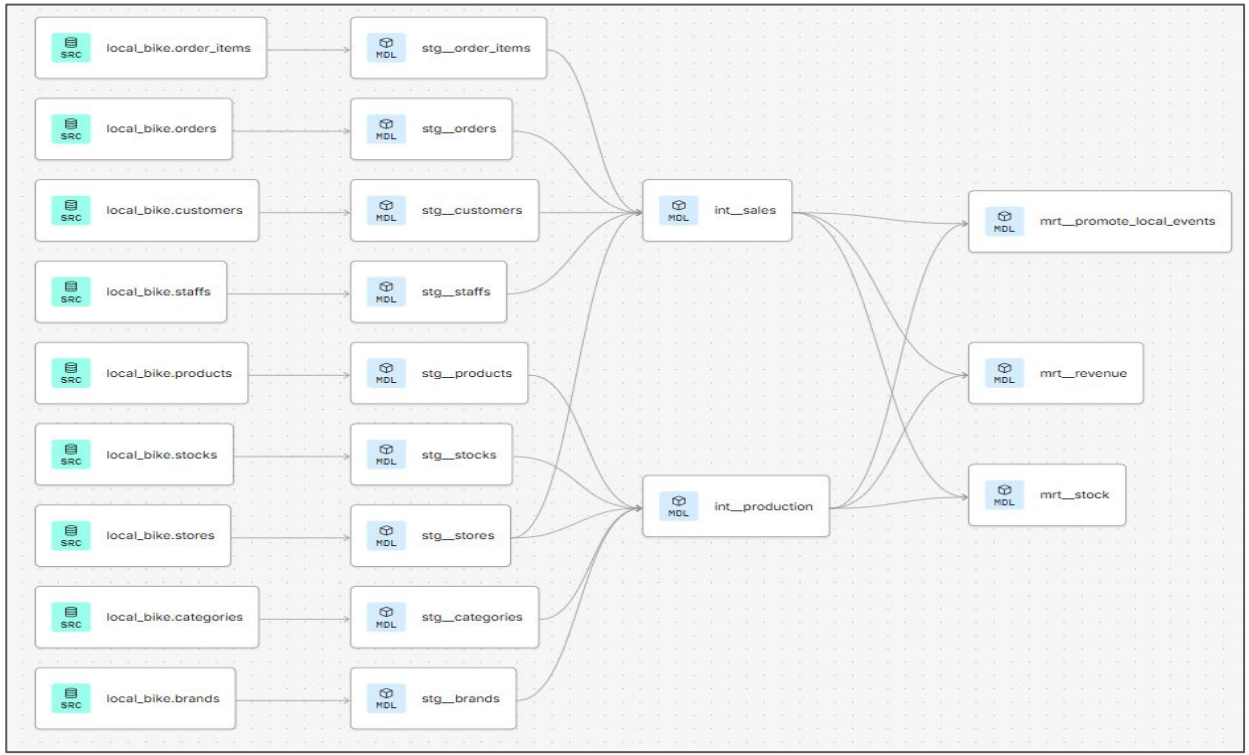
`mrt__revenue` revenue by store, category, brand
`mrt__stock` stock and sales history
`mrt__promote_local_events` top 10 cities

Quality and documentation

Staging	`schema.yml` with model and column descriptions and data types. Each primary key carries `meta: primary_key`, a uniqueness test and a `not_null` test. Each foreign key is checked by a `relationships` test.
Intermediate and mart	Descriptions written in a `.md` file, referenced from one `.yml` file per model.
dbt_project.yml	`persist_docs` enabled and materialization defined per layer.
README.md	Project objectives and conventions.

(dbt) Lineage

Sources: schema local_bike



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(dbt) Lineage: improvements

`stg__stores` feeds both intermediate models.

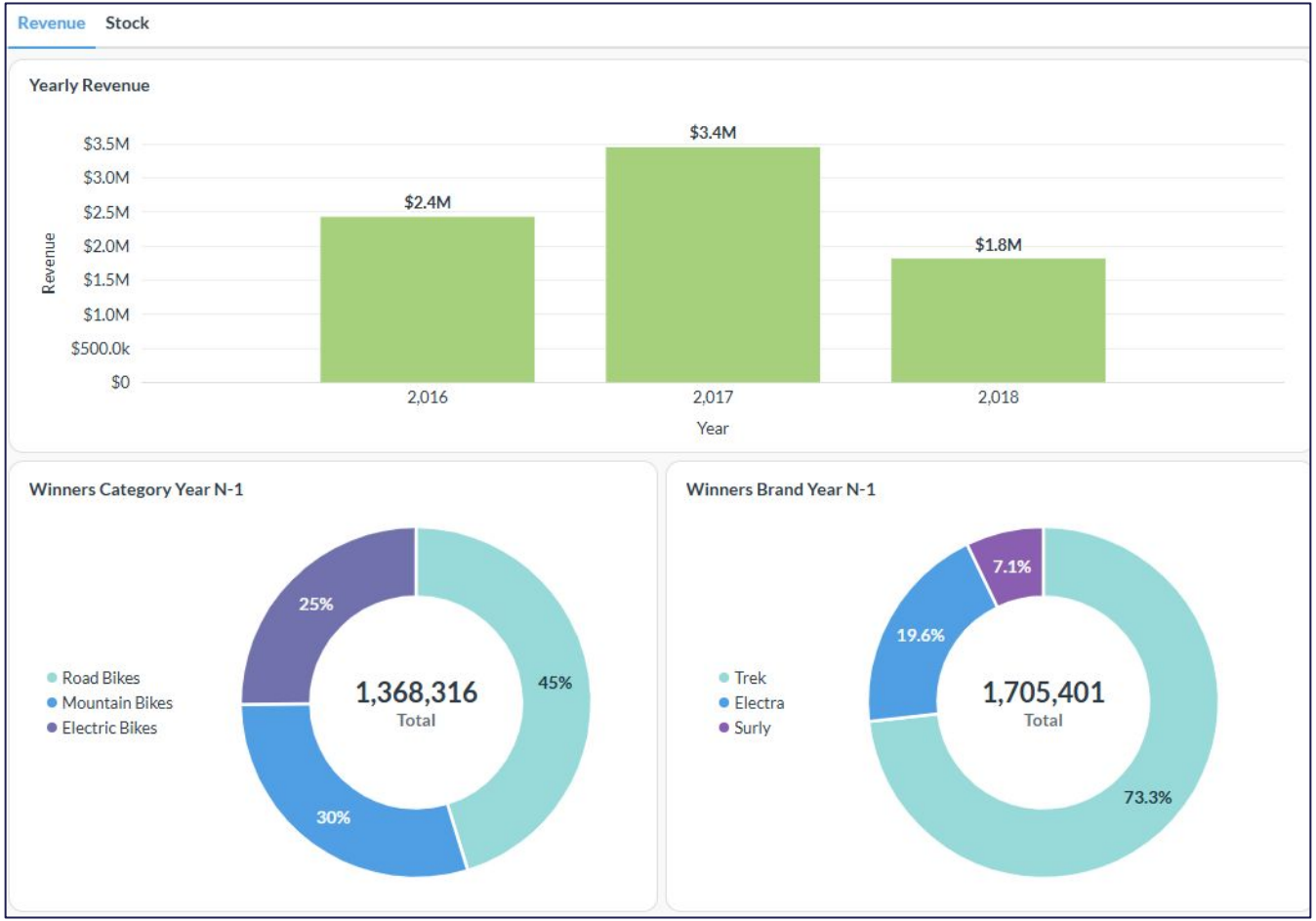
Both intermediate models feed `mrt__revenue` and `mrt__promote_local_events` : sales come from `int__sales`, category and brand from `int__production`.

Possible solution: create a single intermediate model `int__products` (products, category, brand, without stock).

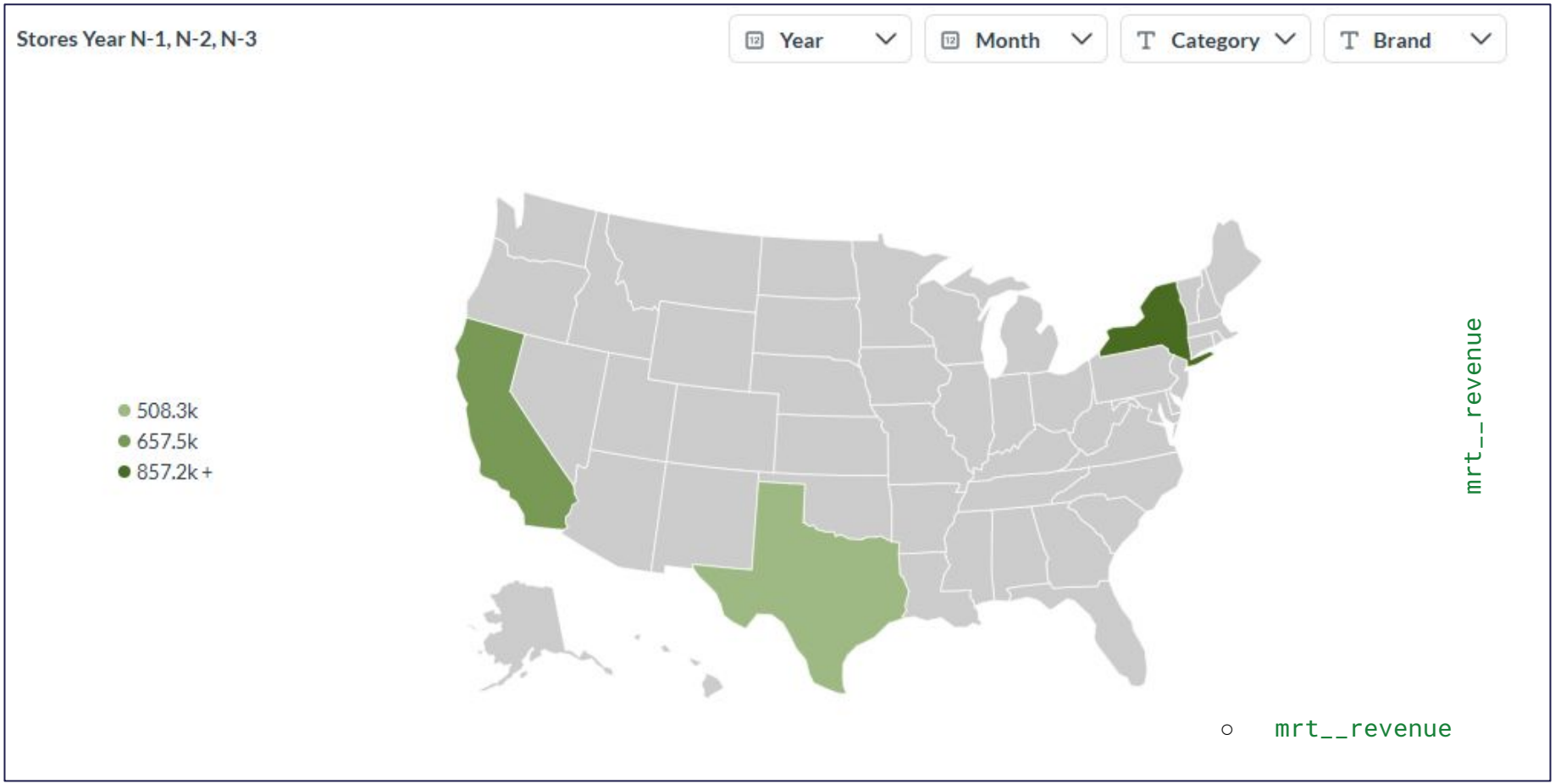
4. Visualization and sharing

https://github.com/isamerch/local_bike

mrt__revenue



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Interactive US map in the dashboard, with filters for Year, Month, Category and Brand.

Stock status

`mrt__stock`

Store	Prd	Prd Name	Model Year	Stock Qty	Stock Amount	Category	Brand	Last Sold	Total Sold
CA	12	Electra Townie ...	2016	16	8,800	Cruisers Bicycles	Electra	April 18, 2018	29
NY	12	Electra Townie ...	2016	2	1,100	Cruisers Bicycles	Electra	October 26, 2017	105
TX	12	Electra Townie ...	2016	30	16,500	Cruisers Bicycles	Electra	April 15, 2018	19
TX	13	Electra Cruiser ...	2016	19	5,130	Cruisers Bicycles	Electra	August 22, 2017	24
NY	13	Electra Cruiser ...	2016	1	270	Cruisers Bicycles	Electra	January 15, 2018	107
CA	13	Electra Cruiser ...	2016	13	3,510	Cruisers Bicycles	Electra	January 4, 2018	26
NY	14	Electra Girl's H...	2016	18	4,860	Cruisers Bicycles	Electra	March 18, 2018	88
TX	14	Electra Girl's H...	2016	4	1,080	Cruisers Bicycles	Electra	June 17, 2017	11
CA	14	Electra Girl's H...	2016	8	2,160	Cruisers Bicycles	Electra	April 21, 2018	25
CA	15	Electra Moto 1 ...	2016	3	1,590	Cruisers Bicycles	Electra	April 23, 2018	50
TX	15	Electra Moto 1 ...	2016	22	11,660	Cruisers Bicycles	Electra	October 27, 2017	10

Self-service business rules

`mrt__stock`

Business rules can be added directly in the editor.

Example

Create a calculated column that identifies rows where stock equals 0 and the last sale was less than 6 months ago, and automatically assigns them the status "Restock". Then color those rows and add a filter based on this condition so they are easy to spot.

Top 10 cities: Winners Year N-1

mrt__promote_local_events

State	City	Total Sold	Top Cat	Top Brand
CA	Oxnard	90	Children Bicycles	Trek
NY	Scarsdale	81	Cruisers Bicycles	Electra
CA	San Jose	81	Children Bicycles	Electra
NY	Mount Vernon	78	Mountain Bikes	Trek
TX	Houston	75	Cruisers Bicycles	Electra
NY	Baldwinsville	69	Road Bikes	Trek
NY	Amityville	63	Road Bikes	Trek
NY	Liverpool	63	Children Bicycles	Electra
NY	Monroe	63	Cruisers Bicycles	Electra
CA	Palos Verdes Peninsula	63	Mountain Bikes	Trek